

# Compact Optical Mirrors with Broadband Polarization-Insensitive Reflectivity using Coupled Photonic Crystal Slabs

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**Abstract.** It was recently demonstrated that a photonic crystal slab can function as a mirror for externally incident light with near-complete reflectivity over a broad wavelength range for transverse magnetic (*TM*) polarized light. We propose a coupled photonic crystal structure that enables broadband reflection over a sizeable angular range for both transverse electric (*TE*) and *TM* polarized light. We expect such mirror to play important roles in optical devices where polarization independent high reflectivity is needed.

**Index Terms:** Guided Resonance, Optical Filters, Sensors, Photonic Crystals.

**D**IELECTRIC optical mirrors with near 100% reflectivity play important roles in optical devices including filters, sensors, switches, and laser structures. The conventional way of obtaining high-reflectivity dielectric mirrors for applications including optical Micro-Electrical-Mechanical-Systems (MEMS) is to use multi-layer films, since metallic mirrors are extremely lossy at optical frequencies. Also, in optical MEMS research, reducing the displacement of mechanical tuning is of great interest since such reduction directly increases the response speed and decreases the actuation force. However, the multilayer nature makes it hard to combine these devices with MEMS technology for sensors and tunable filters applications, since up to 100 dielectric layers is often required for extremely high reflectivity.

In this paper, we demonstrate that broad-band near 100% reflection for normally incident light can be obtained by using a two dielectric layers of photonic crystal slabs. The spectral function of the proposed structure provides a wide angular range of reflection for both *TE* and *TM* polarizations, which is of practical importance when dealing with finite beam sizes. When the beam size is finite, unlike plane waves there are a range of angular components in the incident beam and therefore it is crucial that high reflection for a wide range of angular components is created for various applications.

A photonic crystal slab, which is a building block for our structure, consists of a square lattice of air holes in a high-dielectric film as shown in Figure 1. The operating mechanism of the proposed structure is a guided resonance [1]–[3]. A guided resonance mode in a photonic crystal slab is excited by externally incident light. A typical spatial distribution of the power density in electric fields for a guided resonance mode is shown in Figure 2.

From Figure 2(b), which shows the vertical slice of the power distribution, we can clearly see that the majority of the power is concentrated inside the photonic crystal slab. Although the resonances are strongly confined within the slab, the periodic index contrast provides the phase matching mechanisms that allow these modes to couple into radiation modes and possess a finite lifetime. The interaction between these guided resonance modes and the external radiation gives rise to interesting reflection and transmission properties. The resonance frequency and the

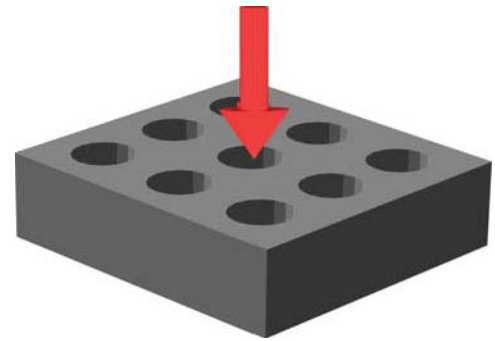


Fig. 1. Schematic of a photonic crystal slab consisting of a square lattice of air holes in a high-dielectric film. The arrow represents the direction of the incident light.

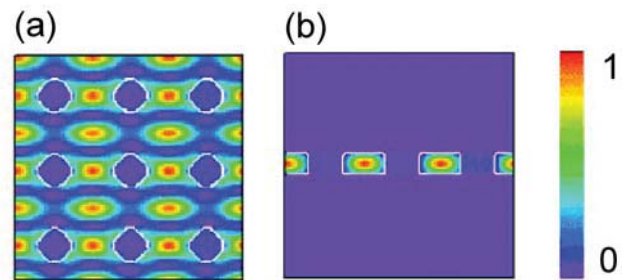


Fig. 2. Spatial distribution of the power in electric fields on (a) a horizontal slice and (b) a vertical slice for normally incident light.

lifetime of the guided resonance are determined by the structure of the photonic crystal, and therefore we can design the desired reflection and transmission properties by engineering the photonic crystal structure, i.e., the radius of the airholes, the thickness of the photonic crystal slabs and the dielectric constant of the photonic crystal slab.

It has recently been shown that when in a photonic crystal slab with dielectric constant of 12, a thickness of  $0.55a$ , and radius of air holes of  $0.4a$ , where  $a$  is the lattice constant, we can obtain a broadband reflection of over 99% [4]. Also, it has been reported that the filter function has a wide angular dependency for *TM* polarization near  $0.53(c/a)$  [5]. However, broadband reflection with a wide angular range is not created

for  $TE$  polarized light near  $0.53(c/a)$  for this structure. For  $TE$  polarization, near 100% reflection with a wide angular range is rather created near  $0.45(c/a)$ . Therefore, if we design the second slab to have near 100% reflection with a wide angular range near  $0.53(c/a)$  for  $TE$  polarization, we would be able to create near 100% reflection for a wide angular range for both  $TE$  and  $TM$  polarized light by coupling two different photonic crystal slabs that compensate each other. This can be theoretically shown using the transfer-matrix formalism. When we define  $r$  and  $t$  as the reflection and transmission coefficient of a single photonic crystal slab, the transfer matrix  $T$  for a single slab reduces to the following form [6]:

$$T = \begin{pmatrix} t + \frac{rr^*}{t^*} & -\frac{r^*}{t^*} \\ -\frac{r}{t} & \frac{1}{t} \end{pmatrix} = \begin{pmatrix} \frac{1}{t^*} & -\frac{r^*}{t^*} \\ -\frac{r}{t} & \frac{1}{t} \end{pmatrix}. \quad (1)$$

When there are two photonic crystal slabs separated by a displacement of  $h$ , the total transfer matrix becomes:

$$T_{total} = \begin{pmatrix} \frac{e^{-j\phi}}{t_1^* t_2^*} + \frac{r_2 r_1^* e^{+j\phi}}{t_2^* t_1^*} & -\frac{r_2^* e^{-j\phi}}{t_1^* t_2^*} - \frac{r_1^* e^{+j\phi}}{t_2^* t_1^*} \\ -\frac{r_1 e^{-j\phi}}{t_1 t_2^*} - \frac{r_2 e^{+j\phi}}{t_1 t_2} & \frac{r_1 r_2^* e^{-j\phi}}{t_1 t_2^*} + \frac{e^{+j\phi}}{t_1 t_2} \end{pmatrix}, \quad (2)$$

where  $\phi = \frac{\omega}{c}h$  is the phase shift that the wave acquires as it propagates through the air cavity that two photonic crystal slabs are forming. Also,  $r_1$ ,  $t_1$  and  $r_2$ ,  $t_2$  are the reflection and transmission coefficients of the first and second photonic crystal slabs, respectively. From the transfer matrix of the total system, we can extract the total transmitted power density of the coupled slab structure as follows:

$$|t_{total}|^2 = \frac{|t_1 t_2|^2}{|r_1 r_2|^2 + 2|r_1 r_2| \cos \left\{ \arg\left(\frac{r_2}{r_1 t_2^*}\right) \right\} + 1}. \quad (3)$$

From Eq. (3), we can clearly see that when the transmission through either of the two slabs is zero, the total transmission is zero and hence we obtain near 100% reflection.

As a physical realization of the theoretical analysis, using a first principles finite-difference time-domain (FDTD) simulation [7], we consider the two slab structure schematically shown in Figure 3. The first slab has a dielectric constant of 12, radius of  $0.4a$  and thickness of  $0.55a$ . From Figure 4, we can see that near frequency  $0.45(c/a)$ , where we have the first resonant reflection, the reflection spectrum has a wide angular range of reflection for  $TE$  polarized light, whereas near frequency  $0.53(c/a)$ , where we have the second resonant reflection, the reflection spectrum has a wide angular range of reflection for  $TM$  polarized light.

Using this information, we design the second slab such that it would completely compensate for the angular dependency of the reflectivity in the first slab. This can be done by shifting the first resonance, which creates a wide angular range of reflection for  $TE$  polarized light, to frequency  $0.53(c/a)$ . By choosing a radius of  $0.35a$  and a thickness of  $0.2a$ , with the same dielectric constant as the first slab, we obtain a wide angular range of reflection for  $TE$  polarized light near frequency  $0.53(c/a)$ . When we couple these two different photonic crystal slabs, we indeed create near 100% reflection for a wide angular range for both  $TE$  and  $TM$  polarized light as shown in Figure 5. Two different slabs compensate each

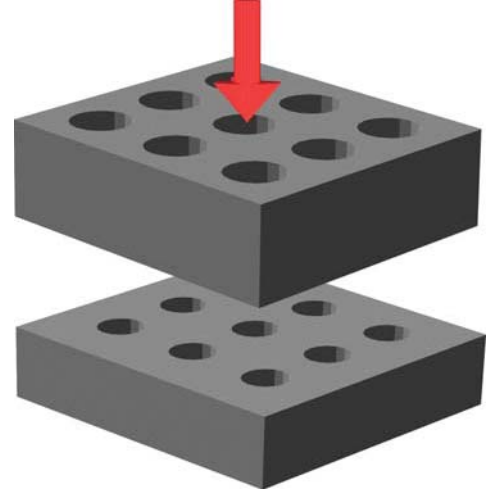


Fig. 3. Schematic of a photonic crystal filter consisting of two coupled photonic crystal slabs. The arrow represents the direction of the incident light. The first slab has a dielectric constant of 12, radius of  $0.4a$  and thickness of  $0.55a$ . On the other hand, the second slab has the same dielectric constant as the first slab, but has a radius of  $0.35a$  and a thickness of  $0.2a$ .

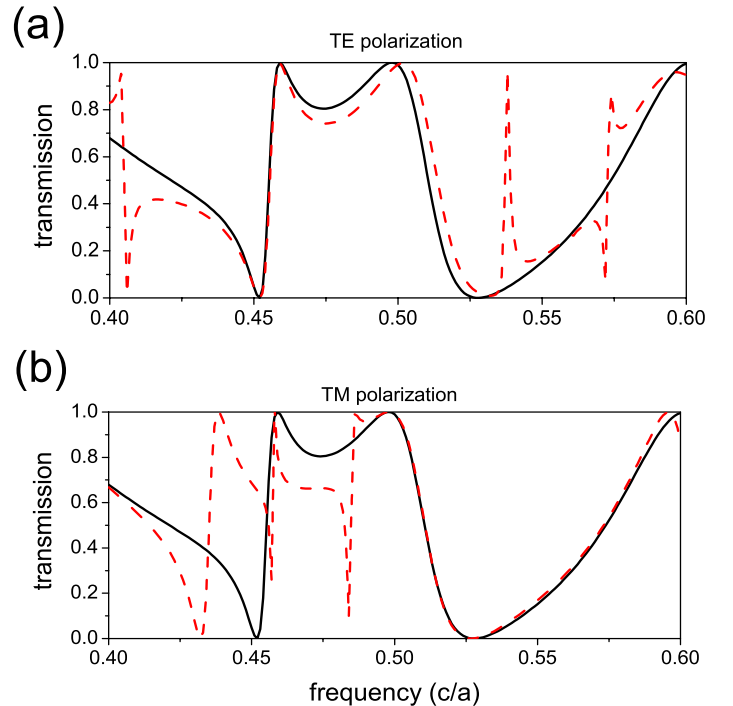


Fig. 4. Transmission spectrum for various incident angles through a photonic crystal slab with dielectric constant of 12, radius  $0.4a$  where  $a$  is the lattice constant, and thickness  $0.55a$ . (a) shows the transmission when incident light has  $TE$  polarization and (b) shows the transmission when incident light has  $TM$  polarization. Both in (a) and (b), solid line is the transmission upon normally incident light, while the dashed line is with incident angle  $1^\circ$  near frequency  $0.53(c/a)$ .

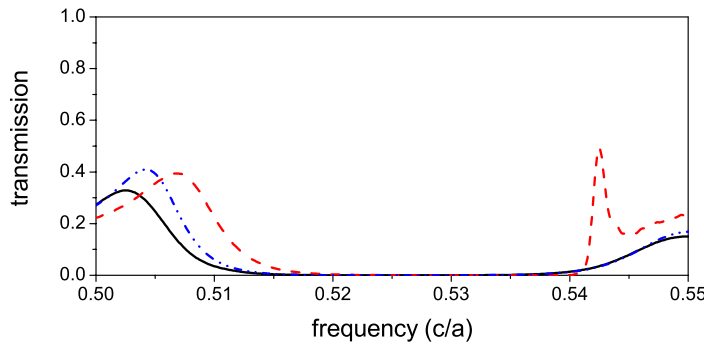


Fig. 5. Transmission spectrum for coupled photonic crystal slabs. The first slab has a dielectric constant of 12, a radius of  $0.4a$ , where  $a$  is the lattice constant, and a thickness of  $0.55a$ . The second slab has the same dielectric constant as the first slab, with a radius  $0.35a$  and thickness of  $0.2a$ . The solid line is for normal incident light, the dashed line is for  $TM$  polarization, and the dash-dotted line is for  $TE$  polarization with incident angle  $11^\circ$  near frequency  $0.53(c/a)$ .

other to give near 100% reflection for an angular range up to  $11^\circ$  for both  $TE$  and  $TM$  polarized light as predicted in the theory.

As final remarks, the photonic crystal slab mirror, which consists of only two single dielectric layers, is far more compact than traditional multi-layer film structures commonly used, where the use of up to 100 dielectric layers is often required to accomplish a Q-factor of a few thousands with a desired line shape. In addition, such a structure represents an attractive application of photonic crystal structures, as the structure functions with a relatively large optical aperture, and does not suffer from the insertion loss problems that often limit the practical use of 2-D photonic crystal waveguides. We therefore expect these novel and compact devices to be useful in optical communication systems.

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